

Hence, we can see that as KE increases, λ decreases, and it will not be linear.

30 Ans: D

$$p = h/\lambda = hf/c$$

$$F = N\Delta p / t = 2Nhf / tc, \text{ where } N = \text{number of photons}$$

$$\text{Since intensity} = (N/t)(hf)(1/A) \quad \Rightarrow \quad Nhf/t = IA$$

$$\text{Hence, } F = 2IA/c$$